

WATER LEVEL DETECTION PROJECT

TEAM 9

Prepared for

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Abstract

This report, prepared for Aliaxis (client contact: Nigel Greig), presents the design and evaluation of an analogue electronic system for measuring water level in a water tank. The system comprises a capacitive probe, a transmitter, and a receiver. PVC was selected as the probe dielectric material, achieving a measurement accuracy of 97.51% at 10% water level. The transmitter encodes water level information by modulating an 850 nm IR light signal, which is decoded by the receiver and represented as a stable DC output voltage. With our current design we are able to transmit IR light at a range of 2.3m from transmitter to the receiver. The report details the investigation and optimisation of the probe, the design rationale for both the transmitter and receiver, and an evaluation of system performance in terms of range, accuracy, and stability. Recommendations for a suitable battery and solar charging system are also provided. The system was validated through simulation using FEMM and LTspice, and a physical prototype was constructed and tested. Key findings include a measured output voltage range of 0.95 V to 2.76 V, stable output at full tank conditions, and increased signal noise at low water levels. A 30 Ah LiFePO₄ battery supported by a minimum 0.83 m² solar panel is recommended for seven-day autonomous operation.

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1 Introduction

1.1 Background

Farms across New Zealand rely on consistent water supply, yet rainfall varies significantly between regions and seasons. Water tanks are therefore essential for storing rainwater during periods of drought or low rainfall. However, without an effective monitoring system, farmers risk unexpected shortages, wasted water, and inefficient use of storage capacity. This project, commissioned by Aliaxis, develops a proof-of-concept analogue system that detects water level in a tank (of height 2500mm) via a capacitive probe, transmits this data optically to a nearby receiver, and outputs a stable DC voltage representing the current water level. The system is designed with scalability in mind for real-world farm deployment. This report details the probe design, transmitter and receiver design, system performance, and recommendations for battery and solar charging.

1.2 Client requirements

The client has specified seven key requirements guiding this design. System accuracy must be validated through Finite Element Analysis (FEA). Data transmission must occur without RF emissions, avoiding the need for certification against standards such as CE, FCC, and RCM. Appropriate housing solutions to mitigate RF emissions should also be considered conceptually. Similarly, electronic subsystems to address RF conductivity and susceptibility should be proposed at schematic level, without requiring full implementation at the PoC stage. The measurement output must be sufficiently stable to serve as a reliable input for external PID control systems. The system must achieve adequate transmission range for PoC demonstration, with a clear pathway for future range expansion. Finally, given solar-charged battery operation, power consumption must be carefully managed to ensure sustainable deployment.

1.3 Design tools and methodology

Four tools and approaches underpinned the development process. The software FEMM was used for electromagnetic field simulation, (Meeker D, 2019) enabling capacitance modelling across water levels and informing probe material selection. LTspice provided circuit-level simulation of the transmitter and receiver, allowing

design optimisation prior to physical implementation. Prototyping validated simulation results against real-world behaviour, identifying discrepancies between theoretical and measured performance. Finally, a top-down design approach was adopted, decomposing the system from client requirements into functional subsystems: probe, transmitter, and receiver, each designed and optimised independently before integration.

2 Water level probe

The water level probe operates on the principle of capacitive sensing. As illustrated in Figure [1], the probe is constructed as a cylindrical capacitor with 4 inner cavities, comprising an inner copper conductor, a PVC dielectric sheath, and an outer copper shell.

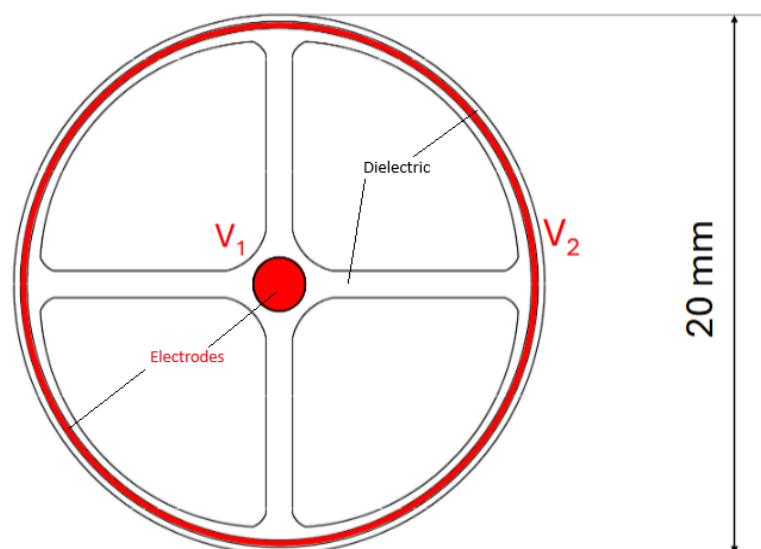


Figure [1]: Probe design and geometry

When the probe is submerged, water fills the cavities within the probe. Since water has a significantly higher relative permittivity than air (approximately 80 vs 1), the total effective permittivity of the medium within the probe increases as more of its length is submerged, resulting in a higher measured capacitance. This provides a direct and predictable relationship between water level and capacitance, which forms the basis of the measurement. The capacitance at each water level was determined using FEMM finite element simulation, with results used to evaluate probe accuracy and inform material selection. This capacitance variation is converted into a voltage

signal by the transmitter circuit for optical transmission to the receiver. The probe spans a measurement range of 0 mm to 2500 mm of tank water level.

2.1 Optimised Configuration

The primary design decision for the probe was the selection of the dielectric sheathing material, for which PVC and silicone rubber (SR) were evaluated across four criteria: measurement stability, durability and lifetime, environmental impact, and ease of installation. PVC was selected as the preferred material due to several key advantages it held over silicone rubber. Its higher dielectric constant yields a capacitance range of 143 pF to 1347 pF, compared to 87 pF to 588 pF for silicone rubber (see Figure [4] in Appendix C). The larger magnitude of capacitance values mean that parasitic capacitances, which are unavoidable in any real circuit, represent a smaller fraction of the total measured signal, resulting in improved measurement stability. PVC also offers a well-documented service life exceeding 100 years under standard operating conditions (Folkman, 2021), making it well suited to a long-term outdoor installation. While silicone rubber offers superior low-temperature flexibility (remaining pliable to approximately -60°C vs. -5°C for PVC), this advantage is not critical for the intended New Zealand deployment environment. From an installation perspective, PVC's rigidity requires careful alignment during fitting and adequate sealing, but once installed it maintains a stable fixed position, which is beneficial for measurement consistency. The material comparison is summarised in Table C.1 (see Appendix C for full analysis and Table C.1).

2.2 Accuracy

Probe accuracy was assessed using FEMM simulation results. PVC exhibited a greater capacitance range than silicone rubber (143 pF to 1347 pF versus 87 pF to 588 pF); this indicates a smaller minimum detectable height change and therefore greater resolution. As shown in Figure [5], this advantage is most pronounced at low water levels, which represent the most critical operating condition if we assume a farmer is more concerned about a tank running dry than overflowing. PVC was therefore selected on the basis of superior accuracy at low fill levels. Full sensitivity calculations are provided in Appendix C.

3 System Electronics

3.1 Transmitter

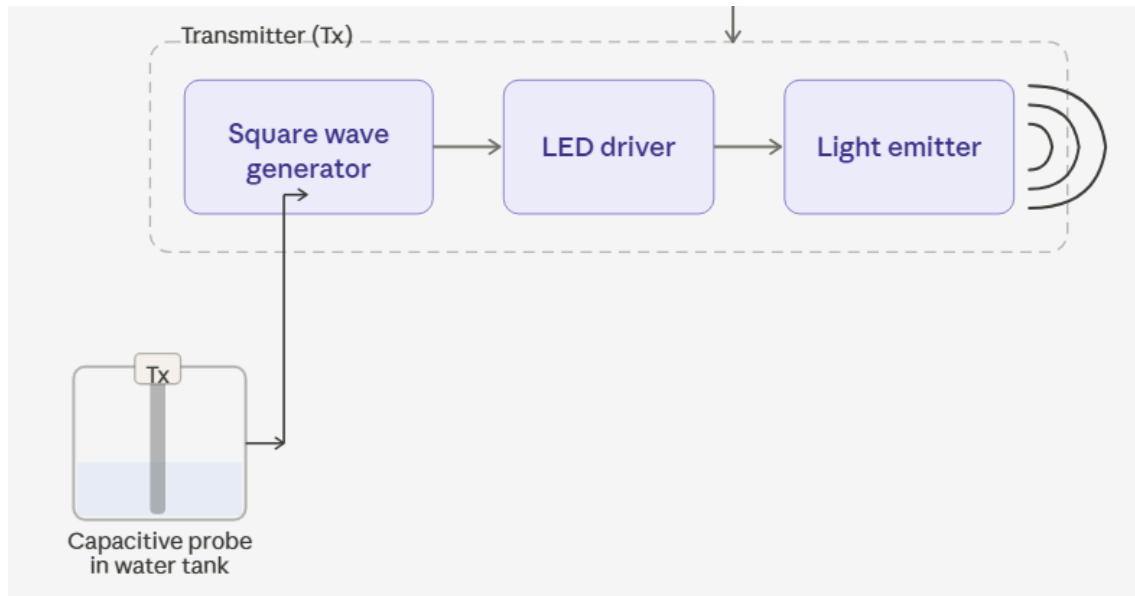


Figure [2] : Transmitter block diagram

Details of the transmitter block diagram are provided in Appendix D

3.1.1 Minimization of signal loss

To address signal loss, a torch reflector was mounted around the IR LED to focus the emitted beam toward the receiver, reducing divergence losses. For driving the LED, MOSFET switching produces clean, fast pulse edges, making the signal easier for the receiver comparator to distinguish from noise. Also, operating at 850 nm (OSRAM, n.d.-b) provides inherent resilience against ambient visible light interference, improving the optical link's signal-to-noise ratio across varying environmental conditions.

3.1.2 Maximisation of range

Range was maximised through four compounding decisions: the 30% duty cycle allows an increased peak LED current beyond what 50% duty would allow at equivalent average power; the SFH4557's high current rating and 850 nm emission match the photodiode's peak sensitivity, maximising the optical link budget; the torch reflector reduces beam divergence; and the LM7805 maintains consistent LED drive current as the battery discharges. (OSRAM, n.d.-a; OSRAM, n.d.-b) Future improvements could include focusing lenses, higher-power LEDs, or a laser source.

3.2 Receiver

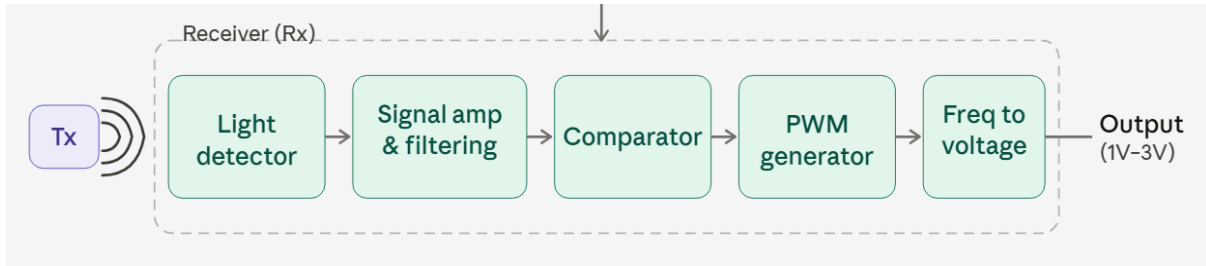


Figure [3]: Receiver block diagram

Details of the receiver block diagram are provided in Appendix E

3.2.1 Minimization of signal loss

Considering the limits of our design, there are three scenarios where our signal may be lost; when there are obstacles which physically obstruct the path between our transmitter and receiver, when the receiver is situated outside of the feasible range, or when the two stages are misaligned. In these cases where there is no signal received, the receiver outputs a distinct voltage of 3V which lies outside of our 0.95V - 2.76V range - this is a clear indicator for signal loss.

Outside of these cases, a torch reflector has also been used on the receiver to focus incident light onto the photodiode to further minimise the chance of signal loss occurring.

3.2.2 Maximization of range

A 220 kΩ transimpedance resistor converts the photodiode's output current into a voltage signal, maximising received signal amplitude. A two-stage TLC082 amplification chain provides a total gain of approximately 2420, with a comparator stage converting the amplified sinusoidal signal into a clean square wave, compensating for the bandwidth sacrifice required to achieve sufficient gain within the TLC082's 10 MHz GBW limit (Texas Instruments, 2015). Full amplification stage calculations are provided in Appendix E.

3.3 Signal capture arrangement

The IR LED and torch reflector are positioned along a direct line-of-sight axis toward the receiver photodiode, ensuring maximum optical power delivery. The transmitter is

mounted at the top of the water tank, with the reflector oriented to direct the focused beam toward the fixed receiver location. This arrangement minimises misalignment losses and ensures consistent signal strength across the full operating range.

3.4 Cost Analysis

Component costs were sourced from DigiKey NZ at bulk pricing (per 1000 units), reflecting scalability considerations. The receiver totals NZD \$15.12 and the transmitter NZD \$6.63, giving a combined electronics cost of approximately NZD \$21.75 (Appendix A). The NE555 timer was selected over a TLC082 Schmitt trigger oscillator for transmitter, reducing cost from \$2.88 to \$0.43 per unit while offering superior timing stability. For receiver, the TLC082 op-amp was chosen over cheaper LM358 (\$0.25 per unit) despite its higher price (\$2.88), justified by its significantly higher slew rate (16 V/ μ s vs 0.3 V/ μ s) and gain-bandwidth product (10 MHz vs 0.7 MHz), both essential for faithful signal amplification at 3 - 30 kHz (Texas Instruments, 2015; Texas Instruments, 2024). (Further details are found in Appendix A)

4 Recommendations

4.1 Battery

A 30 Ah capacity is required for seven days of autonomous operation - informed by Wellington's record seven-day overcast period (1News, 2020). LiFePO₄ chemistry was selected for its safe operating temperature range of -20°C to 60°C during discharge (EBL Official, 2025), suitable for NZ conditions where sub-zero temperatures occur only at night. A 12V 30Ah Fruugo lithium battery is recommended at NZD \$74, for being a low-cost and rechargeable option for 12V applications. (Fruugo, n.d.) Full calculations & details are present in Appendix B.

4.2 Solar charging system

We decided that our solar charging system should be able to fully charge our battery within one day of average sunlight. Average solar power in NZ is between 2.6 kWh and 4.2 kWh (Solargis, 2021). We want farms across NZ to be able to use our product reliably so we designed for 2.6 kWh. Efficiency of solar panels also varies from around 15% to 22% (Project Solar UK, 2026) so designing for 15% efficiency allows us to consider the worst case scenario and allows leeway for degradation of

the solar panel over time, ensuring longevity of the system. To charge our 30Ah battery within one day we needed a solar panel of approximately 0.83m^2 . To meet our requirements we recommend the Renogy 175W N-type solar panel (Renogy, n.d.) which costs \$160, has an efficiency of 25%, and is compatible with 12 V systems. This solar panel would last a long time and has dimensions of 1110mm by 764mm so it would fit within the available mounting area on a typical water tank.

4.3 Other recommendations

The system should be housed in a sealed IP-rated enclosure to protect against moisture and dust. Proper grounding, shielding, and filtering will reduce electrical noise. A charge controller with battery protection is recommended to manage solar variability. Future range improvements should consider optical lenses, higher-power LEDs, or laser transmission with precision-polished reflectors.

5 Conclusions

This report has presented the design and evaluation of a proof-of-concept analogue water level sensor for New Zealand farm water tanks, comprising a capacitive probe, transmitter, and receiver, validated through FEMM simulation, LTspice, and physical prototyping. The system meets all client requirements. Infrared optical transmission at 850 nm eliminates RF emissions, removing the need for CE, FCC, or RCM certification. PVC dielectric selection achieves 97.51% measurement accuracy at 10% water level, which is the most critical condition while satisfying the FEA-validated accuracy requirement. A 2.3 metre transmission range was demonstrated, with scalability achievable through higher-power LEDs or laser sources. The receiver output of 0.95 V to 2.76 V, with a distinct 3 V fault signal on transmission loss, supports reliable PID integration. A 30 Ah LiFePO₄ battery with a minimum 0.83 m^2 solar panel meets the battery life requirement under worst-case NZ solar conditions. Total PoC electronics cost is NZD \$21.75 (more detailed information can be found Appendix A)

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Appendix A - Bill of materials

Using digikey.nz and using price per 1000 components to find the cost we found that the receiver total is \$15.11794 and the Transmitter total is \$6.62467 together they cost \$21.74261 NZD or \$21.75 NZD.

Bill of materials for Receiver:

Table A.1

Part type	Model number	Designator	Manufacturer	Quantity	Cost NZD each (for 1000)	Line cost
op-amp	TLC082CP	U2, U3, U5	Texas instruments	3	\$2.87519	\$8.62557
555 Timer	NE555P	U4	Texas instruments	1	\$0.43188	\$0.43188
Regulator	LM78M05CT/ NOPB	U1	Texas instruments	1	\$1.37600	\$1.37600
Photodiode	SHF 203 P	D1	Osram	1	\$0.74624	\$0.74624
Schottky Diode	BAT85S-TAP	D2	Vishay	1	\$0.23509	\$0.23509
HDR2X1	640456-2	CON1, CON2	TE Connectivity AMP Connectors	2	\$0.14909	\$0.29818
Resistor	Several	R2, R3, R4, R5, R6, R7, R8, R9, R10, R11,	Stackpole Electronics	19	\$0.02106	\$0.40014

		R12, R13, R16, R18, R21, R22, R23, R27, R33				
Ceramic Capacitor	Several	C2, C3, C4, C5, C6, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C19, C20, C21, C22, C23, C24, C25, C26	KEMET	23	\$0.12385	\$2.84855
Electrolytic Capacitor	ESH476M06 3AE3AA	C1	KEMET	1	\$0.15529	\$0.15529

Bill of materials for Transmitter:

Table A.2

Part type	Model number	Designator	Manufacturer	Quantity	Cost NZD each (for 1000)	Line cost
555 Timer	NE555P	U2	Texas instruments	1	\$0.43188	\$0.43188
Regulator	LM78M05CT/ NOPB	U1	Texas instruments	1	\$1.37600	\$1.37600
IR LED	SFH 4557	D1	Osram	1	\$0.57234	\$0.57234

Schottky Diode	BAT85S-TAP	D2, D3	Vishay	2	\$0.23509	\$0.47018
N-Channel MOSFET	MTP3055VL	Q1	Onsemi	1	\$1.38753	\$1.38753
HDR2X1	640456-2	CON1	TE Connectivity AMP Connectors	1	\$0.14909	\$0.14909
HDR4X2	TSW-104-07-F-D	P1	Samtec	1	\$0.48048	\$0.48048
Resistor	Several	R1, R2, R3, R4	Stackpole Electronics	4	\$0.02106	\$0.08424
Ceramic Capacitor	Several	C3, C4, C5, C6, C7, C9, C10, C11, C12, C13	KEMET	11	\$0.12385	\$1.36235
Electrolytic capacitor	ESH476M06 3AE3AA	C1, C2	KEMET	2	\$0.15529	\$0.31058

Cost Analysis Detail:

All component costs are sourced from DigiKey NZ at bulk pricing (per 1000 units) to reflect real-world scalability for large-scale farm deployment.

Transmitter: The NE555 timer (\$0.43/unit) was selected over a TLC082 op-amp Schmitt trigger oscillator (\$2.88/unit). Beyond cost savings, the 555 timer offers independent charge/discharge control via two diodes, making duty cycle adjustment more straightforward, and is purpose-built for timing applications - making it both cheaper and more stable than an op-amp equivalent.

Receiver: The TLC082 op-amp (\$2.88/unit) was selected over the LM358 (\$0.25/unit). While more expensive, the TLC082's slew rate of 16 V/ μ s (compared to 0.3 V/ μ s for the LM358) and gain-bandwidth product of 10 MHz (compared to 0.7 MHz) are essential for faithful amplification and comparator switching at the system's operating frequency range of 3 - 30 kHz. The LM358 would introduce unacceptable signal distortion at these frequencies, directly compromising detection accuracy and transmission range. The additional cost is therefore justified by the performance requirements of the design.

Appendix B - Solar energy calculations

Battery calculations:

Our device runs at 12V and uses 0.153A our power is $12V * 0.153A = 1.836W * 24 = 44Wh/12 = 3.67 Ah$ for one day capacity.

To run it for 7 days $= 3.67Ah * 7 = 25.69Ah/0.85 = 30Ah$

Divided by 0.85 for a lithium ion battery to have enough usable charge.

If we were using lead-acid batteries, divide by 0.5 to have enough usable charge.

$25.69/0.5 = 51.38Ah$

Solar panel calculations:

Lowest average solar power available in NZ is 2.6 kWh and solar panels have varying efficiency of 15% to 24% to simulate worst case scenario we used 15%

$2.6 kWh * 0.15 = 0.39 kWh$ the amount of solar energy within 1 square metre we can capture.

To account for some energy loss during conversion we added 5% more energy

$44Wh * 1.05 = 46.2Wh$

$0.0462 kWh/0.39 kWh = 0.1184m^2$ to charge a days worth of energy, to charge 7 days worth $0.1184 * 7 = 0.83m^2$

Dimensions of the Renogy 175W solar panel are 1110mm x 764mm $= 0.848m^2$ which is enough for our purposes

Battery Selection:

Two 12V 30Ah battery options were evaluated against the calculated 30Ah requirement. The comparison is summarised in Table B.1.

Table B.1 Battery option comparison

	Vicedeal 12V 30.6Ah LiFePO4	Fruugo 12V 30Ah Li-ion
Chemistry	LiFePO4	Li-ion
Capacity	30.6Ah	30Ah
Cycle Life	5000+ cycles	2000+ cycles
BMS	Built-in 30A BMS	Built-in (overcharge, overdischarge, overcurrent, short circuit)
Max Discharge Current	30A	5A continuous, 10A peak
Weight	1.05 kg	<300 g
Certifications	CE, KC, RoHS, UL	CE, FCC, RoHS, UL, KC
Cost	NZD \$190	NZD \$74

The Fruugo 12V 30Ah lithium ion battery is recommended for the proof-of-concept stage. Its 30Ah capacity precisely meets the calculated requirement, it carries multiple safety certifications, has 2000+ cycles, and at NZD \$74 it is significantly cheaper than the Vicedeal option. Its weight of under 300g also makes it practical for mounting on a water tank. (Fruugo, n.d.)

The Vicedeal LiFePO4 battery offers superior cycle life (Vicedeal, n.d.) (5000+ vs unspecified) and a higher continuous discharge current of 30A, making it the stronger long-term choice for a production deployment. However, at approximately NZD \$190 it carries a higher cost that is difficult to justify at the PoC stage. For a future scaled deployment where longevity is critical, the POWORTH battery would be the preferred recommendation.

Appendix C

	PVC	Silicon
Cost	Low (Cost-effective)	High
Stability	Good	Excellent
Lifetime and Durability	Good	Excellent
Ease of Installation	Excellent	Good
Carbon Footprint	High	Low
Temperature Performance	Good	Excellent
Accuracy	Excellent at 100%, good at 10%	Excellent at 100%, low at 10%
Sensitivity	Good	Low

Table C.1. Shows differences between PVC and Silicon Rubber

Probe:

Stability

System stability is strongly influenced by the dielectric material's resistance to parasitic capacitance effects. As the inherent capacitance of the sensing structure increases, the relative impact of parasitic capacitances is reduced, leading to improved measurement stability. Based on the simulation results presented in Figure [4], the capacitance range for a PVC probe (143 pF to 1347 pF) is consistently higher than that of a silicone rubber (SR) probe (87 pF to 588 pF). Consequently, the median capacitance value of PVC is significantly greater, indicating that PVC is less susceptible to parasitic capacitance interference. This suggests that, from a stability perspective, PVC provides more reliable and robust performance under the simulated conditions.

Lifetime and Durability

The long-term performance of the dielectric material in an aqueous environment is a critical design consideration. Given that the system operates in relatively clean water, established empirical data can be used to evaluate material durability. According to industry reports (e.g., PIPA), PVC pipes can achieve a service life exceeding 100

years under typical operating conditions, depending on installation quality and environmental factors (Folkman, 2021) . In contrast, silicone rubber typically offers a service life of over 10 years before significant material degradation occurs (Elastostar, n.d.) .

Temperature resilience is another important factor. At low temperatures, materials may become brittle and prone to cracking. PVC exhibits embrittlement within a temperature range of approximately -5°C to -40°C , depending on its formulation. Silicone rubber, however, demonstrates superior low-temperature performance, remaining flexible down to approximately -60°C to -70°C (Elastostar, n.d.) . Therefore, while PVC offers exceptional longevity under standard conditions, silicone rubber provides better reliability in extreme cold environments.

Carbon Footprint

Environmental impact is assessed through the material's carbon footprint, which accounts for greenhouse gas emissions across its lifecycle, including production, use, and disposal. PVC generally has a higher environmental impact due to its derivation from fossil fuels and the release of harmful by-products, such as vinyl chloride monomer and dioxins, during manufacturing (GreenMatch, 2024) .

In contrast, silicone rubber is produced from silica and, although its manufacturing process is energy-intensive, it typically results in fewer toxic emissions (Xiao & Zhang, 2023) . Furthermore, its chemical stability and longer functional lifespan in certain applications can reduce the frequency of replacement, thereby lowering its overall lifecycle environmental impact (Silicones Europe, 2019) . As a result, silicone rubber is often considered the more environmentally sustainable option despite its higher initial energy cost.

Probe Accuracy Calculations

The following calculations determine the minimum detectable height change (Δh) and theoretical accuracy for both PVC and silicone rubber (SR) dielectric options across all simulated water levels. The reliable minimum detectable capacitance change is fixed at $\Delta C = 3 \text{ pF}$.

Tank geometry:

$$\text{Volume} = \pi \times (1.75)^2 \times 2.5 = 24.05 \text{ m}^3$$

PVC - Capacitance range:

$$C_{\text{empty}} = 143 \text{ pF}, C_{\text{full}} = 1347 \text{ pF}$$

$$C_{\text{full}} - C_{\text{empty}} = 1347 - 143 = 1204 \text{ pF}$$

 Δh :

$$\Delta h = (3 / 1204) \times 2500 = 6.23 \text{ mm}$$

Sample calculation at 10% water level (PVC):

$$V_{10\%} = 10\% \times 24.05 = 2.405 \text{ m}^3$$

$$V_{\Delta h} = \pi \times (1.75)^2 \times 0.00623 = 0.0599 \text{ m}^3$$

$$\text{Accuracy} = (1 - (0.0599 / 2.405)) \times 100\% = 97.51\%$$

SR - Capacitance range:

$$C_{\text{empty}} = 87 \text{ pF}, C_{\text{full}} = 588.52 \text{ pF}$$

$$C_{\text{full}} - C_{\text{empty}} = 588.52 - 87 = 501.52 \text{ pF}$$

$$\Delta h = (3 / 501.52) \times 2500 = 14.95 \text{ mm}$$

Results for all water levels:*Table C.2 - PVC accuracy vs water level*

Water Level (%)	Capacitance (pF)	Accuracy (%)
10	262	97.51
20	384	98.75
30	504	99.17
40	625	99.38
50	745.3	99.50
60	865	99.58
70	987	99.64
80	1109	99.69
90	1230	99.72
100	1347	99.75

Table C.3 - Silicone rubber accuracy vs water level

Water Level (%)	Capacitance (pF)	Accuracy (%)
-----------------	------------------	--------------

10	138	94.01
20	188	97.00
30	238	98.00
40	288	98.50
50	338	98.80
60	388	99.00
70	438	99.14
80	488	99.25
90	538	99.33
100	588.52	99.40

Accuracy (%) vs Water Level (%)

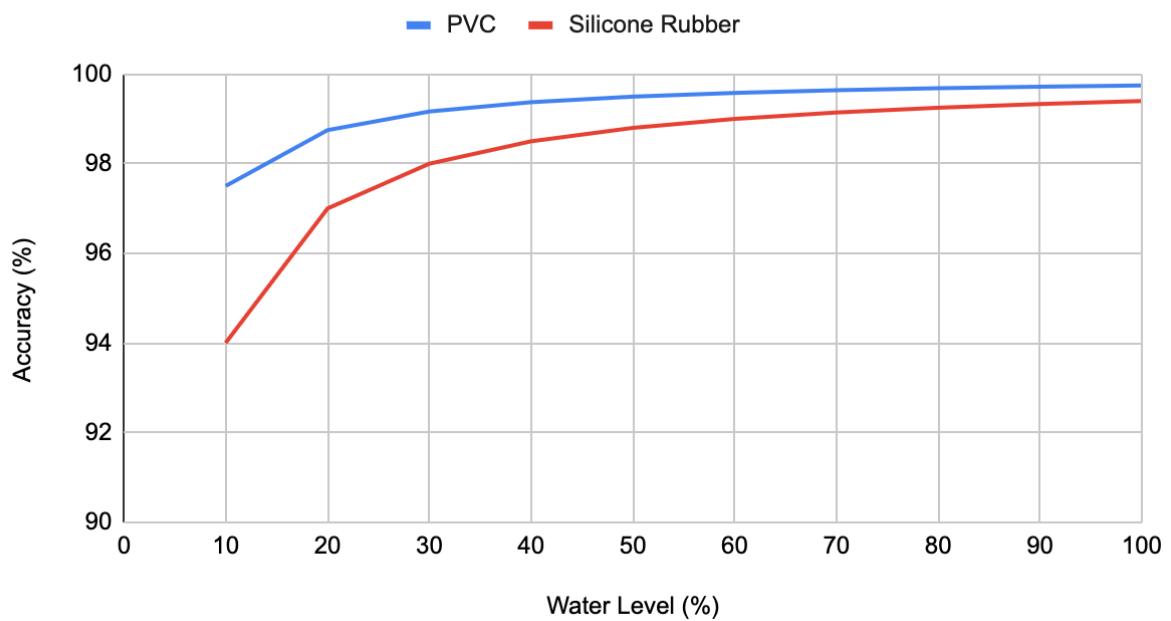


Figure [4] Compares Accuracy vs Water level for PVC and Silicone Rubber

Capacitance vs Water Level

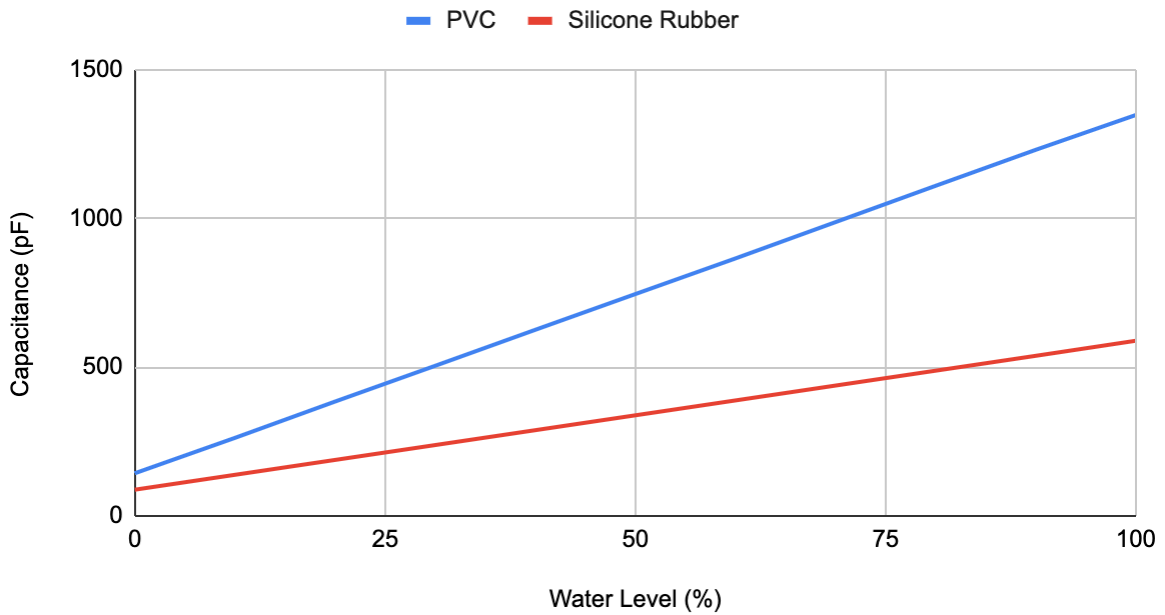


Figure [5] Compares Capacitance vs Water level for PVC and Silicone Rubber

Appendix D

Transmitter:

Transmitter Circuitry

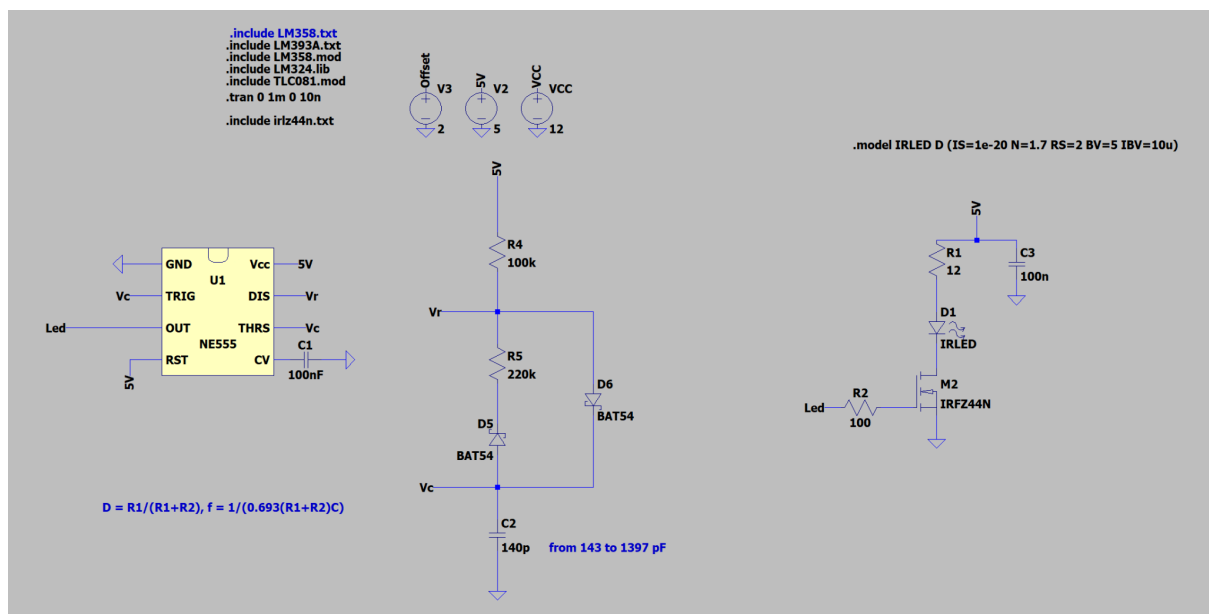


Figure [6] : Full transmitter LTspice schematic

We have designed this transmitter to create 30% duty cycle signals using 555 timer, and allow high peak current to go through our IR led.

Calculations for 30% duty cycle and R2 and R1 value

Duty cycle, $D = R4/(R4+R5)$, ratio needs to match 0.3. With trial and error the best e12 values that gets close to 30 duty cycles is using $R4 = 100k$ ohms and $R5 = 220k$ ohms. To find a suitable value for $R1$ we used 8.2 ohms, so $V_{ref} - V_{led} / R1 = 400mA$, to drive IR led with more power for a short period of time. $V_{ref} = 5V$, $V_{led} = 1.5V$, $R1 = 8.2$ ohms, closest e12 value to get $400mA$ of current, and around $80mA$ of average current through IRled (simulated in LTspice). For our designs, we also prioritised using resistors in the kilo-ohm range rather than ohms or megaohms.

Kilo-ohm values provide the best balance between current consumption, noise performance, and circuit stability. Very low resistances (Ω) draw unnecessarily high current and waste power, while very high resistances ($M\Omega$) introduce noise, leakage errors, and timing instability, especially in circuits such as the 555 timer. Using $k\Omega$ values ensures balanced noise, minimise power loss, and reliable timing ratios across the system.

OPAMP-based Frequency Modulation Approach

A frequency modulation (FM) approach using a triangle-wave-generating integrator fed by a comparator was initially explored to convert changes in capacitance into a measurable signal. The **TLC082 operational amplifier** was selected due to its dual op-amp configuration, which aligns with the PCB constraints and eliminates the need for separate comparator and amplifier components.

The TLC082 operates within a voltage range of approximately $0.18 V$ to $4.3 V$, making it suitable for low-voltage applications. A **DC offset of $2 V$** was introduced to centre the signal within the usable voltage range of the op-amp, ensuring symmetric operation. Additionally, a **hysteresis band of $\pm 1.5 V$** was implemented around this offset to maximise signal swing and improve noise immunity, resulting in a more stable output waveform.

The op-amp's **bandwidth of $10 MHz$** and **slew rate of approximately $16 V/\mu s$** enable operation at relatively high frequencies while maintaining waveform integrity.

A higher operating frequency is desirable, as it increases measurement resolution, allowing small variations in capacitance to be more easily distinguished. Lower-performance alternatives, such as the LM358, were not selected due to insufficient slew rate, which would limit signal fidelity at higher frequencies.

From a cost perspective, the TLC082 is also advantageous, with a unit price of approximately \$2.63 per 1000 units, compared to higher-performance alternatives such as the LTC6221 (\$5.38 per 1000), whose additional bandwidth is unnecessary for this application.

However, a key limitation arises from the **internal parasitic capacitance** of the op-amp, estimated to be around 20 pF at 10 kHz. When the sensing capacitance is relatively small (e.g., 150 pF), this introduces significant error in the frequency response. Although the effect becomes less pronounced at higher capacitance values (e.g., 1500 pF), increasing the operating frequency further amplifies parasitic effects, resulting in substantial measurement inaccuracies. Simulation results indicate that while performance at 10 kHz is acceptable, higher frequencies (e.g., 50 kHz) lead to increased deviation from expected values. This trade-off limits the effectiveness of the opamp-based FM approach for precise measurements.

LED Driver

At our 30% duty cycle we are putting approximately 220mA through our IR LED, a 555 timer can output 200mA at max output. We are above this limit and so decided to use a MOSFET to drive current through the LED. We also considered using a darlington pair to drive the current but the MOSFET was more reliable overall as it had lower power usage, shorter switching times, lower voltage drop and used less components which made it run better and easier to design for. Using the MOSFET driver means the amplitude of current through the LED is controlled by R1 in Figure [6]. $(V_{cc} - V_d - V_m) / R1 = I_p$

Summary

Overall, both frequency modulation and PWM approaches were evaluated. While the FM method offers higher theoretical resolution, it is significantly affected by parasitic capacitance and high-frequency limitations. In contrast, the PWM approach using a

555 timer provides a simpler, more reliable, and cost-effective solution, making it more suitable for the Proof-of-Concept implementation.

Appendix E

Receiver:

Full receiver circuitry:

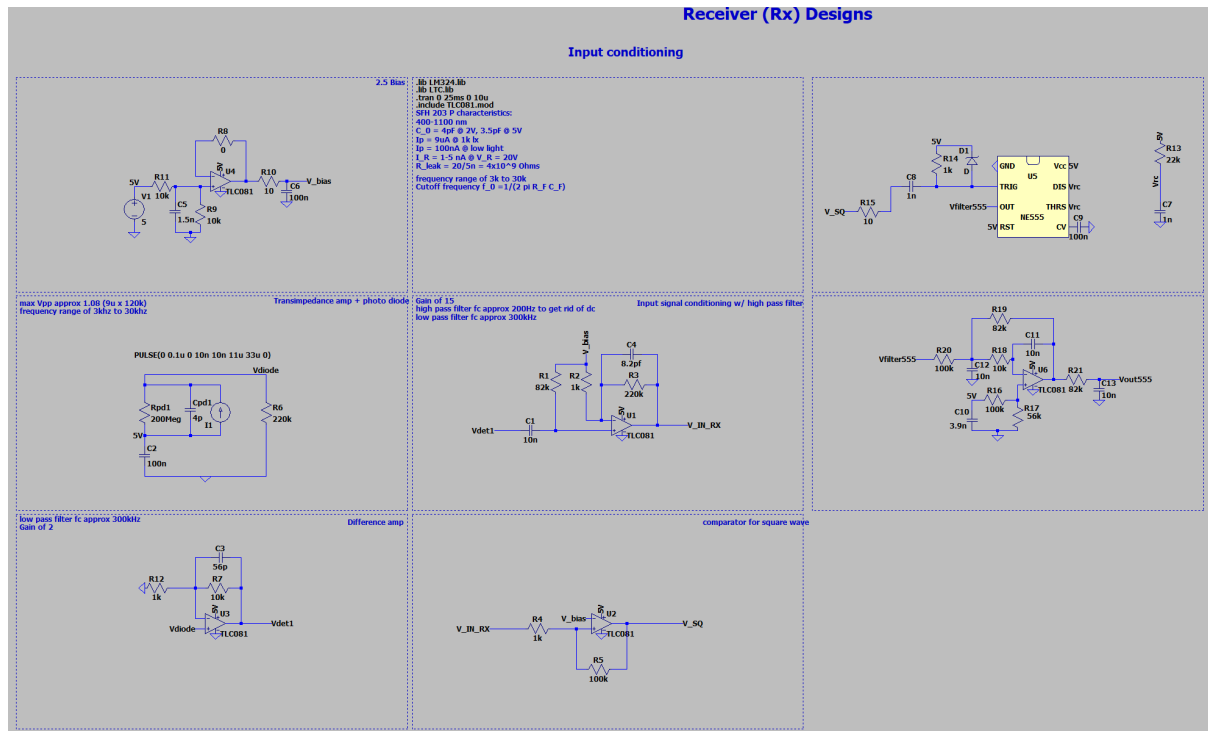


Figure [7] : Full receiver LTspice schematic

We designed the receiver to take the light signal emitting from the transmitter and through a series of stages turn it into a dc voltage corresponding to the frequency of the light signal.

Light detection

To detect light we used a photodiode as it provides a linear relationship with light intensity and has a fast switching time. We designed the Light detecting stage to capture the light signal and turn it into a voltage signal we settled on 220 kilo ohm resistor for this as when we tested a 330 kilo ohm resistor it created a low pass filter with the photodiodes internal capacitance which made us not get a signal but only a dc voltage. We also reverse biased the photodiode against 5V to decrease its

internal capacitance which increases switching speeds and allows for higher resistance values.

The circuit acts as a transimpedance amplifier, converting current into a voltage signal with the given equation:

$$V = IR$$

Signal Amplification and filtering

A two-stage non-inverting amplifier topology was implemented to achieve sufficient signal gain for downstream processing. The circuit schematic is shown in Figure [8]. For a non-inverting amplifier, gain is given by:

$$G = 1 + R_f/R_{in}$$

The first stage was designed for a gain of 11, achieved using $R_f = 10\text{ k}\Omega$ and $R_{in} = 1\text{ k}\Omega$:

$$G_1 = 1 + 10\text{k}/1\text{k} = 11$$

A moderate gain was chosen for the first stage to avoid amplifying the significant noise present in the incoming signal. A high-pass filter with a cutoff of 200 Hz was incorporated to remove low-frequency and DC offset noise, and a low-pass filter with a cutoff of 100 kHz was included to attenuate higher frequency interference.

The second stage was designed for a gain of 221, using $R_f = 220\text{ k}\Omega$ and $R_{in} = 1\text{ k}\Omega$:

$$G_2 = 1 + 220\text{k}/1\text{k} = 221$$

The gain-bandwidth product (GBW) of the TLC082 op-amp (10 MHz) imposed an upper limit on achievable gain, as higher gain reduces the usable bandwidth according to:

$$\text{Bandwidth} = \text{GBW} / G$$

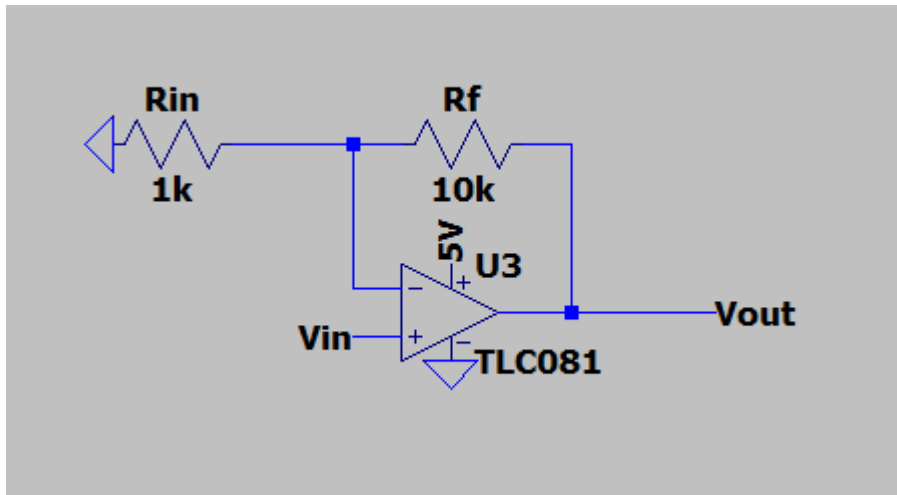


Figure [8]: A non-inverting amplifier circuit, showing component values for gain

Comparator

The input signal to the 555 timer is derived from the output of a comparator, as shown in Figure [9], rather than directly from the amplification stage. This design choice ensures that the input waveform is a clean, well-defined square wave, which improves the consistency and reliability of pulse generation.

Using the comparator output also ensures that the signal maintains a constant amplitude, regardless of variations in light intensity or transmission conditions. This is particularly important in optical transmission systems, where signal attenuation may occur. By standardising the input waveform before it reaches the 555 timer, the system achieves improved noise immunity and more stable operation.

The hysteresis range we used was 100mv which is enough so slight noise won't affect our signal while still giving the benefits of a comparator

The hysteresis voltage is given by:

$$V_{\text{hys}} = 2(R_{\text{in}}/R_{\text{f}}) \times V_{\text{cc}}$$

With $R_{\text{in}} = 1 \text{ k}\Omega$, $R_{\text{f}} = 100 \text{ k}\Omega$, and $V_{\text{cc}} = 5 \text{ V}$:

$$V_{\text{hys}} = 2 \times (1\text{k}/100\text{k}) \times 5 = 0.1 \text{ V} = 100 \text{ mV}$$

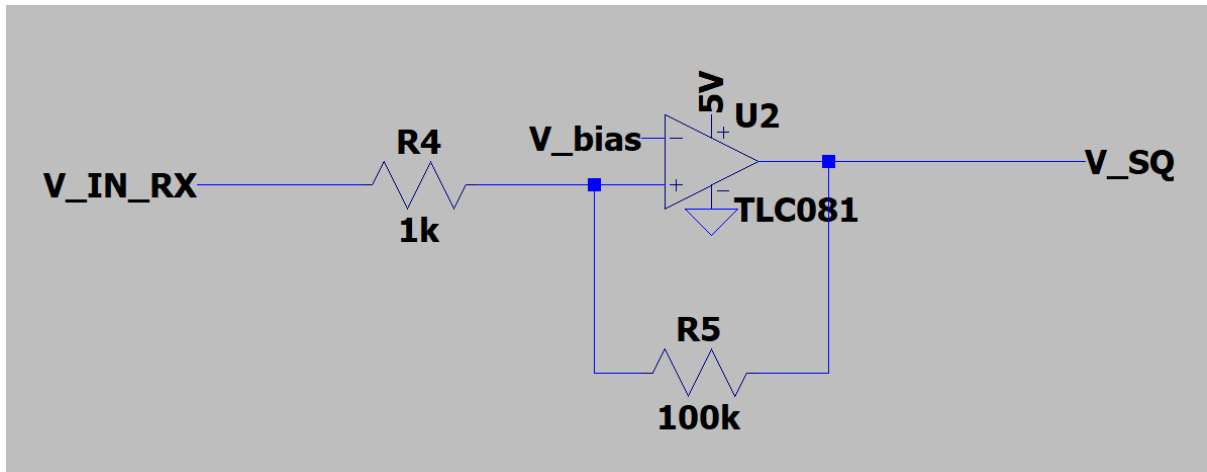


Figure [9]: Comparator circuit with hysteresis.

Offset

We designed an offset using an opamp to give us a stable dc voltage of 2.5V. We made the input resistance low to avoid loading the opamp and used low pass filters to keep the signal a stable dc voltage. We needed this offset so that the amplifying stage and the comparator have a reference voltage and the full signal could be seen as we opted for single rail powering of the opamps for a simpler and easier design.

Pulse Generation

A 555 timer IC was selected for pulse generation due to its ability to produce stable output pulses that are allowed to be larger than the period of which the input signal is high for.

In contrast, the op-amp and diode based pulse generation method produced output signals with average amplitudes in the order of tens of millivolts, which are significantly smaller and more susceptible to noise and component variation. The larger average signal magnitude produced by the 555 timer improves robustness and reduces sensitivity to variations in component values.

As a result, the subsequent filtering stage becomes less dependent on precise component matching. This allows the use of standard E12 series components without significantly affecting performance, simplifying both design and implementation while reducing cost.

The pulse width of the 555 timer is given by:

$$t = 1.1 \times R \times C$$

The circuit schematic is shown in Figure [10]. With $R = 22 \text{ k}\Omega$ and $C = 1 \text{ nF}$:

$$t = 1.1 \times 22\text{k} \times 1\text{n} = 24.2 \text{ }\mu\text{s}$$

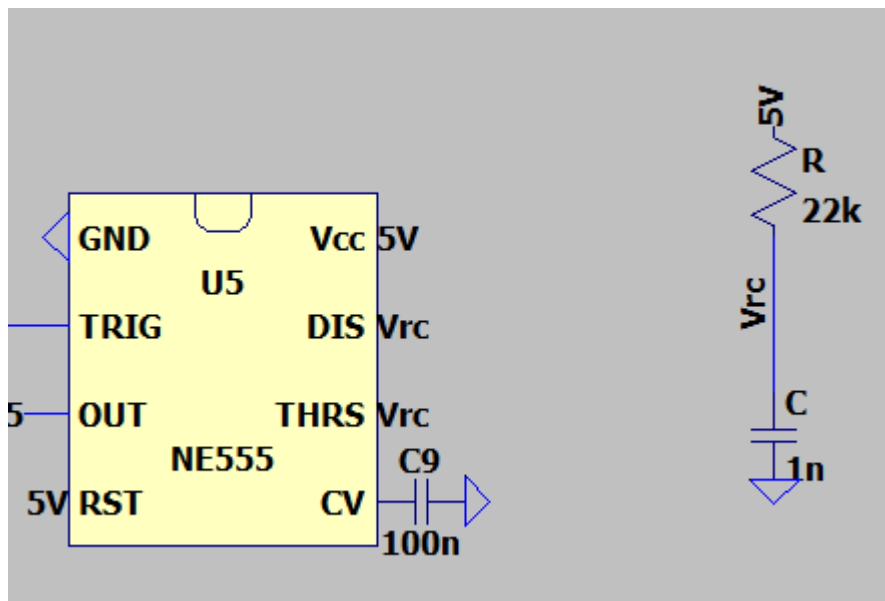


Figure [10]: 555 timer circuit showing timing resistor and capacitor values used to achieve a pulse width of $24.2 \text{ }\mu\text{s}$.

Multiple Feedback (MFB) Filter

A Multiple Feedback (MFB) topology was employed to perform signal conditioning, including amplification, inversion, and filtering, as shown in Figure[11]. This is required to get our signal into our range of 0.95V to 2.76V , depending on the frequency of the signal.

The MFB circuit operates by first applying a negative gain, inverting the signal while scaling its amplitude. We needed it to be an inverting configuration as we wanted to make our low water level of the tank correspond with low voltage level and high water level to high voltage. A DC offset is then applied to shift the conditioned signal into the desired voltage range.

An additional advantage of the MFB configuration is its inherent filtering capability. The topology includes two low-pass filtering stages within the op-amp structure, which help attenuate high-frequency noise and ripple. Furthermore, an additional low-pass filter is placed at the output to smooth the signal further, producing a stable DC level suitable for downstream processing or control applications.

This combination of gain, inversion, and filtering in a single stage improves overall system efficiency and reduces circuit complexity.

The output voltage is given by:

$$V_{\text{out}} = -(R_3/R_1) \times V_{\text{in}} + (1 + R_3/R_1) \times V_{\text{ref}}$$

Where V_{ref} is set by a resistor divider:

$$V_{\text{ref}} = 5 \times R_5/(R_4 + R_5)$$

LTspice Design

The circuit was first designed in LTspice using simulated average input voltages of $V_{\text{in}} = 3.7 \text{ V}$ (empty tank) and $V_{\text{in}} = 0.364 \text{ V}$ (full tank). Setting the desired outputs of 0.95 V and 2.76 V respectively, the following system of equations was solved for $k = R_3/R_1$ and V_{ref} :

$$0.95 = -k(3.7) + (1+k)V_{\text{ref}} \dots (1)$$

$$2.76 = -k(0.364) + (1+k)V_{\text{ref}} \dots (2)$$

Subtracting (1) from (2):

$$1.81 = k(3.336)$$

$$k = 0.543$$

Substituting into (1):

$$V_{\text{ref}} = 1.918 \text{ V}$$

The closest E12 resistor values giving $k \approx 0.543$ were selected as $R1 = 680 \text{ k}\Omega$ and $R3 = 390 \text{ k}\Omega$:

$$k = 390\text{k}/680\text{k} = 0.574$$

V_{ref} was achieved using $R4 = 330 \text{ k}\Omega$ and $R5 = 220 \text{ k}\Omega$:

$$V_{\text{ref}} = 5 \times 220\text{k}/(330\text{k} + 220\text{k}) = 2.0 \text{ V}$$

PCB Implementation

During PCB testing, the measured average input voltages from the 555 timer output differed from simulation, with $V_{\text{in}} = 2.81 \text{ V}$ (empty tank) and $V_{\text{in}} = 0.336 \text{ V}$ (full tank). The component values were recalculated using the same method:

$$0.95 = -k(2.81) + (1+k)V_{\text{ref}} \dots (1)$$

$$2.76 = -k(0.336) + (1+k)V_{\text{ref}} \dots (2)$$

Subtracting (1) from (2):

$$1.81 = k(2.474)$$

$$k = 0.732$$

Substituting into (1):

$$V_{\text{ref}} = 1.736 \text{ V}$$

The closest E12 resistor values giving $k \approx 0.732$ were selected as $R1 = 100 \text{ k}\Omega$ and $R3 = 82 \text{ k}\Omega$:

$$k = 82\text{k}/100\text{k} = 0.82$$

V_{ref} was achieved using $R4 = 100 \text{ k}\Omega$ and $R5 = 56 \text{ k}\Omega$:

$$V_{\text{ref}} = 5 \times 56\text{k}/(100\text{k} + 56\text{k}) = 1.795 \text{ V}$$

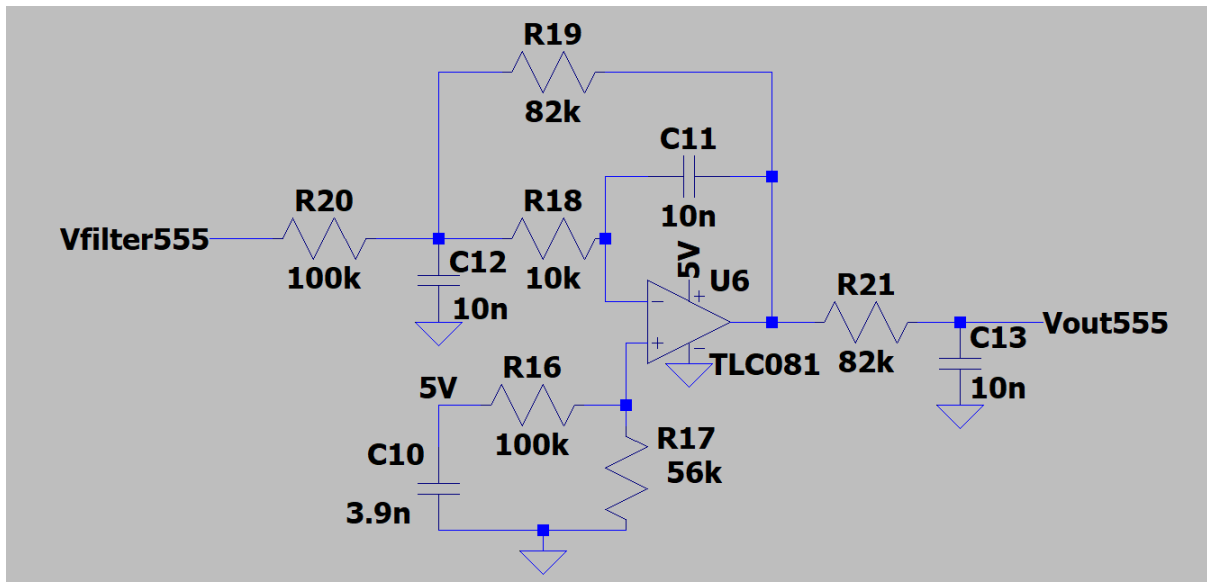
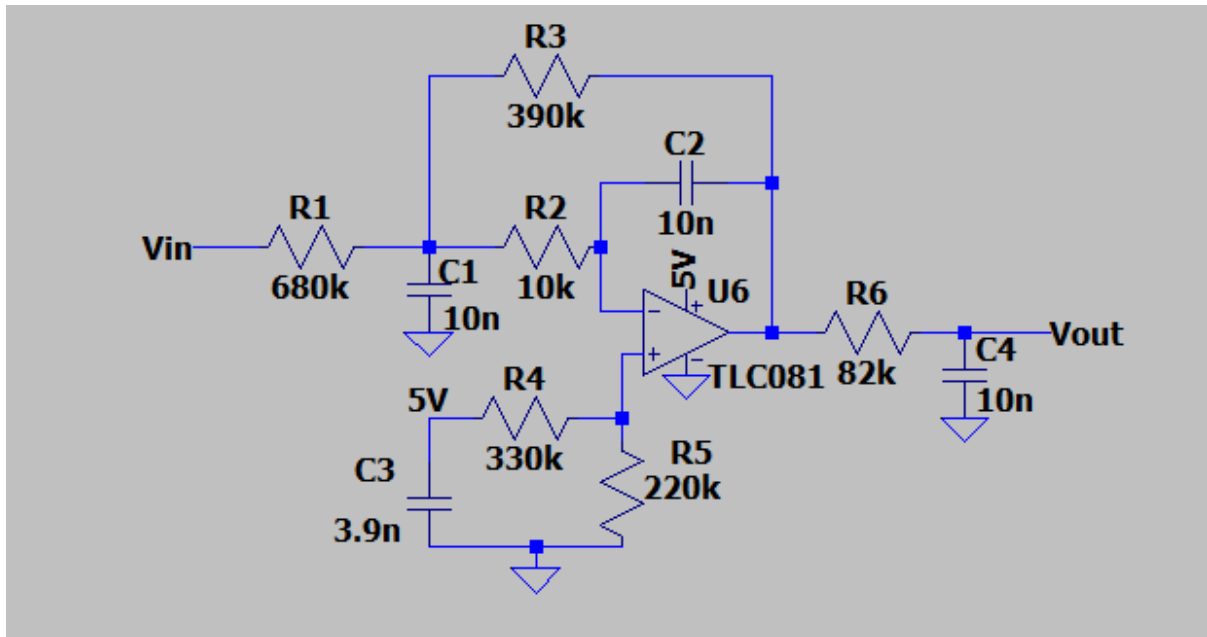


Figure [11]: MFB filter circuit showing component values for gain, inversion, and DC offset stages.

Appendix F

Limitations of Prototype Testing

Several limitations were identified during prototype testing that should be considered when interpreting the results. Component tolerances introduced slight deviations from theoretically calculated values, meaning measured outputs did not perfectly match simulation predictions. Manual alignment between the IR LED and photodiode was required during testing, which is not robust and would be impractical in a real deployment where precise fixed positioning is critical. The prototype used through-hole PCB construction, which simplified debugging but resulted in a bulkier assembly, and repeated soldering during iterative testing caused minor PCB damage that may have affected circuit performance.

Environmental conditions during testing also failed to fully reflect real-world farm deployment. Bright classroom lighting introduced ambient light noise into the photodiode, degrading signal quality beyond what would be expected in a properly shielded outdoor enclosure. Testing was conducted indoors with no exposure to wind, rain, dust, or temperature variation, meaning the system's performance under actual outdoor farm conditions remains unvalidated. Additionally, the prototype was powered by university laboratory power supplies rather than the intended battery and solar charging system, so the effect of supply voltage variation and battery discharge on system performance was not evaluated. These limitations should be addressed in future development stages through outdoor field testing with the recommended battery and solar system.

Future Recommendations

A number of improvements are recommended for future development stages, spanning circuit design, optical performance, environmental robustness, and system scalability.

Circuit and PCB Design

Transitioning from through-hole to surface-mount technology (SMT) components would significantly reduce PCB size and enable automated assembly, lowering manufacturing cost and reducing human soldering errors. SMT construction also reduces parasitic inductance and capacitance, which is beneficial at the system's higher operating frequencies where these parasitics can degrade signal integrity. The 555 timer pulse width should also be more precisely tuned to produce cleaner PWM averaging, improving the accuracy and stability of the DC output voltage. (ALLPCB, 2025)

Optical Performance

Adding collimating lenses in front of the IR LED would concentrate the emitted beam, increasing optical intensity at the receiver and significantly extending transmission range. A corresponding collection lens at the photodiode would increase the area of light captured, improving sensitivity. Larger, precision-polished reflectors on both the transmitter and receiver would further focus the optical path and reduce light scatter. For maximum range, replacing the IR LED with a laser source would provide a significantly narrower beam with far less divergence, enabling transmission over much greater distances. (Technology UK, n.d.) Using multiple LEDs and multiple photodiodes in parallel would increase total optical power and capture area, improving reliability under real-world conditions where minor misalignment or partial obstruction may occur.

Environmental and RF Protection

The photodiode should be shielded from ambient light using an optical bandpass filter matched to the 850 nm operating wavelength, or a physical shroud directing only the transmitted beam toward the sensor. The entire electronics assembly should be enclosed in a sealed IP-rated enclosure to protect against moisture, dust, and temperature extremes encountered in outdoor farm environments. RF susceptibility should be addressed by enclosing sensitive analogue circuitry within a Faraday cage style shielding enclosure, preventing external radio frequency interference from corrupting the low-level photodiode signal. (Wikipedia Faraday Cage, 2026) A charge controller with battery protection features should also be incorporated into the solar charging system to manage solar variability and prevent battery overdischarge.

Validation and Scaling

More specialised testing under New Zealand-specific environmental conditions including temperature extremes, humidity, and direct sunlight, should be conducted to verify system robustness under worst-case operational scenarios. As the system scales toward real-world farm deployment, a full outdoor field trial with the recommended battery and solar system would be necessary to validate long-term performance and reliability.